

OPERATING SYSTEMS (CS304)

Classic Synchronization Problems

Learning Objectives

Understand process synchronization, critical section problems, semaphores, and classic synchronization problems used in operating systems.

Introduction to Process Synchronization

Process synchronization is used to coordinate the execution of multiple processes sharing common resources. It prevents data inconsistency and race conditions.

Critical Section Problem

A critical section is a part of a program where shared resources are accessed. Only one process should execute in the critical section at a time.

Requirements of Critical Section Solution

- Mutual Exclusion
- Progress
- Bounded Waiting

Race Condition

A race condition occurs when multiple processes access and modify shared data simultaneously, producing unpredictable results.

Semaphore

A semaphore is a synchronization tool used to control access to shared resources. It can be binary or counting semaphore.

Producer-Consumer Problem

The Producer generates data and places it into a buffer, while the Consumer removes and processes data from the buffer. Synchronization ensures the buffer does not overflow or underflow.

Producer-Consumer Solution

Semaphores such as Empty, Full, and Mutex are used to coordinate Producer and Consumer activities.

Reader-Writer Problem

Multiple readers can access shared data simultaneously, but writers require exclusive access to prevent inconsistencies.

Reader-Writer Solution

Semaphores and synchronization mechanisms ensure writers are not interrupted and readers access data safely.

Dining Philosophers Problem

Five philosophers sit around a table sharing forks. Synchronization is required to prevent deadlocks and starvation while eating.

Dining Philosophers Solution

Resource ordering, semaphores, and monitors can be used to prevent deadlock situations.

Deadlock and Starvation

Poor synchronization can lead to deadlocks where processes wait indefinitely and starvation where processes never get resources.

Advantages of Synchronization

- Prevents race conditions
- Ensures data consistency
- Supports resource sharing
- Improves system reliability

Exam Important Questions

1. Define Critical Section Problem.
2. Explain Semaphores.
3. Discuss Producer-Consumer Problem.
4. Explain Reader-Writer Problem.
5. Discuss Dining Philosophers Problem.
6. What is Race Condition?

Summary

Synchronization is essential for managing shared resources safely. Classic synchronization problems help understand coordination, deadlock prevention, and resource management in operating systems.